You Should Plug Your Telemed Ultrasound to Mains Power

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Introduction. Ultrasound has been used for speech studies for decades (Gick, 2002; Shawker & Sonies, 1985; Stone, 2005) In recent years Telemed micro ultrasound devices have become something of a norm. While the manual says to use mains power (connect the ultrasound device's power source to a wall socket) it does not say why, and the device will work with just power from the USB-connection to a computer. We discovered by accident why this recommendation exists: When running on USB power the ultrasound device will raise imaging power by steps during each recording. According to Telemed, this is done to avoid setting off a short circuit panic-mode in the USB port. Unfortunately, the stepping of intensity has consequences for data quality. Imaging with lower power produces darker images. In practice this means that at the beginning of the recording the ultrasound image will brighten in steps. This may or may not affect anatomical tracking, and can cause artefacts in analysis metrics such as Pixel Difference (PD) (Palo, 2019) and other analysis methods.

Method and results. We demonstrate the effect with analysis of a technical validation recording of the author. Figure 1 shows the analysis in PATKIT (Palo, Moisik, & Faytak, 2026). As can be seen

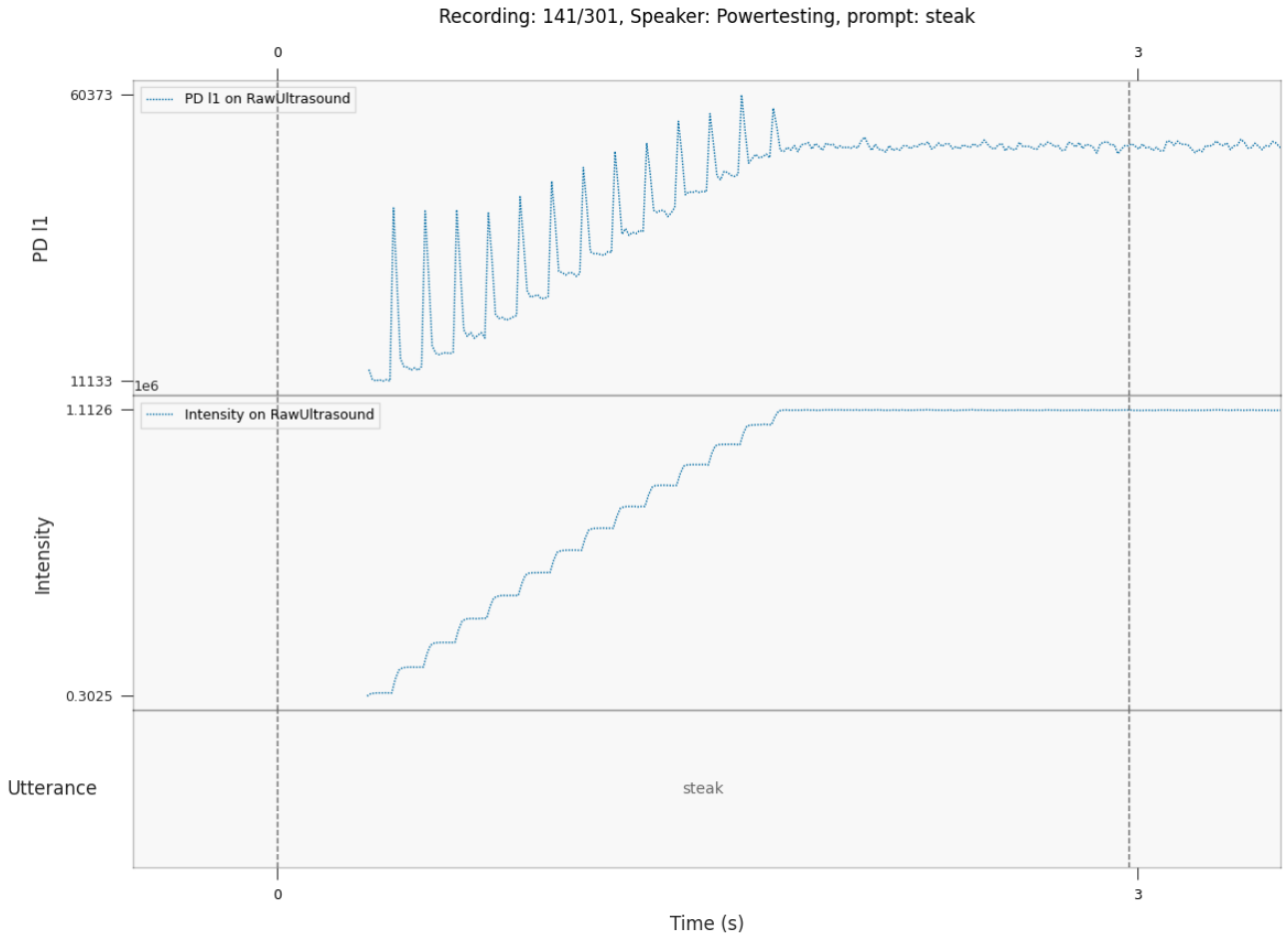


Figure 1: Panels from top to bottom: Pixel Difference (PD), Image overall ultrasound intensity, Sound spectrogram, Sound waveform, and TextGrid (showing only the prompt 'hold still').

the image intensity ramps up by steps from beginning of the ultrasound recording at about $t = 0.3$ s to about $t = 1.73$ s, at which point the intensity plateaus. There is no speech in this recording as the speaker was holding still. There is however a series of spikes in the PD curve (in addition to natural twitching) which correspond to the moments when the intensity rises a step. The effect on image quality can be seen in Figure 2, which shows the frame a the beginning of the recording and a frame at the end of the intensity ramp. The latter image is brighter and has potentially more detail.

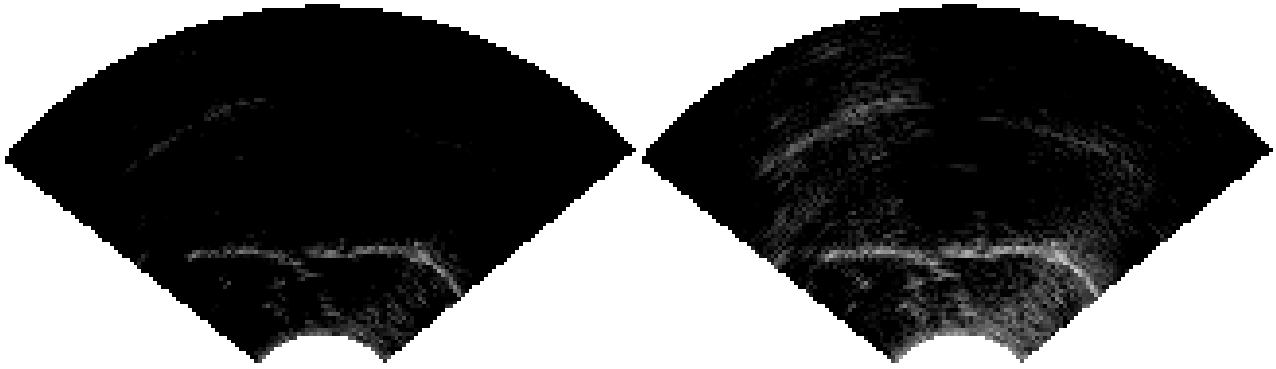


Figure 2: Ultrasound frames at the beginning of the recording (left) and at the end of the intensity ramp (right).

Conclusion. For new data the recommendation is simple: Plug the ultrasound machine into mains power whenever possible and if you need to run it with only USB power, make a note of that in your recording metadata and/or laboratory journal. For old data, if you do not know that the device was plugged in to mains electricity, there are two ways of checking if there is a problem with data quality. First, play a recording and look at the data. If the ultrasound image seems to get brighter they may be a problem. Second, examine the intensity curves in PATKIT. You may need to look at more than one recording to see if intensity is going up step-wise like in Figure 1. In case the intensity is ramping up, any intensity based measures might be unreliable. Examining only PD curves in PATKIT or AAA (Articulate Instruments, 2012) is not reliable as out of 40 test recordings made during the technical testing, only two clearly showed a PD spike artefact pattern seen in Figure 1. Not seeing the pattern in the PD curve of a recording should not be taken as evidence of the absence of the problem in a dataset, nor even the recording in question, as it may affect event timing without being easily detectable.

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